



PREMIUM

Full-Length Mock Test

Mock 01

Computer Science and Information Technology

GATE CS

Total Questions	60
Maximum Marks	108
Duration	180 minutes (3 Hours)
Exam Pattern	MCQ + MSQ + NAT
Negative Marking	MCQ/MSQ: -0.33 per wrong answer
Designed For	EduNeuro Premium

Easy 20%

Moderate 33%

Hard 47%

Based on Previous Year Question Pattern Analysis | Generated by EduNeuro Premium

General Instructions

This question paper contains THREE sections: Section A, Section B, and Section C.

Section A: Multiple Choice Questions (MCQ) — carries one or two marks each. Only ONE option is correct.

Section B: Multiple Select Questions (MSQ) — carries two marks each. ONE OR MORE than one option(s) is/are correct.

Section C: Numerical Answer Type (NAT) — carries one or two marks. Enter the numerical value in the space provided.

Negative Marking: -0.33 marks will be deducted for each wrong MCQ and each wrong MSQ answer.

No negative marking for NAT questions.

There are 60 questions for a total of 100 marks.

Duration: 3 hours (180 minutes).

Use the space provided for rough work. Scribbling on the question paper is permitted.

Do not ask for clarification from the invigilator. Interpret questions as given.

Paper Structure

Section	Type	Questions	Marks/Q	Total Marks	Negative
A	MCQ	35	1 / 2	35	-0.33
B	MSQ	5	2	10	-0.33
C	NAT	20	1 / 2	35	None
Total		60		100	

Marking Scheme

Correct MCQ: +1 mark (or +2 marks as specified per question)

Wrong MCQ: -0.33 marks

Correct MSQ: +2 marks

Wrong MSQ (partial or no match): -0.33 marks

Correct NAT: +1 mark (or +2 marks as specified per question)

No answer / Unattempted: 0 marks

NAT wrong answer: 0 marks (no negative marking)

Section A — Multiple Choice Questions (MCQ)

Q.1 [1 Mark] [EASY] [Algorithms — Graph Algorithms]

Which data structure implements FIFO?

- A) Stack
 - B) Queue
 - C) Tree
 - D) Graph
-

Q.2 [1 Mark] [EASY] [Data Structures — Hashing]

Time complexity of binary search?

- A) $O(n)$
 - B) $O(\log n)$
 - C) $O(1)$
 - D) $O(n^2)$
-

Q.3 [1 Mark] [EASY] [DBMS — Indexing]

Normal form where all non-prime attributes depend on key:

- A) 1NF
 - B) 2NF
 - C) 3NF
 - D) BCNF
-

Q.4 [1 Mark] [EASY] [OS — File Systems]

In deadlock prevention, hold-and-wait is prevented by:

- A) Preemption
 - B) Resource ordering
 - C) Mutual exclusion
 - D) Allocation at once
-

Q.5 [1 Mark] [EASY] [CN — TCP/IP]

Class A IP range:

- A) 0-127
 - B) 128-191
 - C) 192-223
 - D) 224-239
-

Q.6 [1 Mark] [EASY] [TOC — Turing Machines]

A regular grammar generates:

- A) Type 0
 - B) Type 1
 - C) Type 2
 - D) Type 3
-

Q.7 [1 Mark] [EASY] [COA — Addressing Modes]

Cache hit ratio if access time = 10ns, hit time = 1ns, miss penalty = 100ns:

A) 0.1

B) 0.5

C) 0.9

D) 0.99

Q.8 [1 Mark] [EASY] [Digital Logic — Combinational Circuits]

Which is not a page replacement algorithm?

- A) FIFO
 - B) LRU
 - C) Optimal
 - D) FCFS
-

Q.9 [1 Mark] [EASY] [Engineering Mathematics — Linear Algebra]

Which data structure implements FIFO?

- A) Stack
 - B) Queue
 - C) Tree
 - D) Graph
-

Q.10 [1 Mark] [EASY] [Programming — C Programming]

Time complexity of binary search?

- A) $O(n)$
 - B) $O(\log n)$
 - C) $O(1)$
 - D) $O(n^2)$
-

Q.11 [1 Mark] [EASY] [Algorithms — Time Complexity]

Normal form where all non-prime attributes depend on key

- A) 1NF
 - B) 2NF
 - C) 3NF
 - D) BCNF
-

Q.12 [1 Mark] [EASY] [Data Structures — Hashing]

In deadlock prevention, hold-and-wait is prevented by:

- A) Preemption
 - B) Resource ordering
 - C) Mutual exclusion
 - D) Allocation at once
-

Q.13 [2 Marks] [MODERATE] [DBMS — Transactions]

Class A IP range:

- A) 0-127
 - B) 128-191
 - C) 192-223
 - D) 224-239
-

Q.14 [2 Marks] [MODERATE] [OS — CPU Scheduling]

A regular grammar generates:

- A) Type 0

B) Type 1

C) Type 2

D) Type 3

Q.15 [2 Marks] [MODERATE] [CN — Routing]

Cache hit ratio if access time = 10ns, hit time = 1ns, miss penalty = 100ns:

- A) 0.1
 - B) 0.5
 - C) 0.9
 - D) 0.99
-

Q.16 [2 Marks] [MODERATE] [TOC — Grammars]

Which is not a page replacement algorithm?

- A) FIFO
 - B) LRU
 - C) Optimal
 - D) FCFS
-

Q.17 [2 Marks] [MODERATE] [COA — Addressing Modes]

Which data structure implements FIFO?

- A) Stack
 - B) Queue
 - C) Tree
 - D) Graph
-

Q.18 [2 Marks] [MODERATE] [Digital Logic — Minimization]

Time complexity of binary search?

- A) $O(n)$
 - B) $O(\log n)$
 - C) $O(1)$
 - D) $O(n^2)$
-

Q.19 [2 Marks] [MODERATE] [Engineering Mathematics — Calculus]

Normal form where all non-prime attributes depend on key:

- A) 1NF
 - B) 2NF
 - C) 3NF
 - D) BCNF
-

Q.20 [2 Marks] [MODERATE] [Programming — Data Structures]

In deadlock prevention, hold-and-wait is prevented by:

- A) Preemption
 - B) Resource ordering
 - C) Mutual exclusion
 - D) Allocation at once
-

Q.21 [2 Marks] [MODERATE] [Algorithms — Greedy]

Class A IP range:

- A) 0-127

B) 128-191

C) 192-223

D) 224-239

Q.22 [2 Marks] [MODERATE] [Data Structures — Hashing]

A regular grammar generates:

- A) Type 0
 - B) Type 1
 - C) Type 2
 - D) Type 3
-

Q.23 [2 Marks] [MODERATE] [DBMS — Transactions]

Cache hit ratio if access time = 10ns, hit time = 1ns, miss penalty = 100ns:

- A) 0.1
 - B) 0.5
 - C) 0.9
 - D) 0.99
-

Q.24 [2 Marks] [MODERATE] [OS — File Systems]

Which is not a page replacement algorithm?

- A) FIFO
 - B) LRU
 - C) Optimal
 - D) FCFS
-

Q.25 [2 Marks] [MODERATE] [CN — Network Security]

Which data structure implements FIFO?

- A) Stack
 - B) Queue
 - C) Tree
 - D) Graph
-

Q.26 [2 Marks] [MODERATE] [TOC — Decidability]

Time complexity of binary search?

- A) $O(n)$
 - B) $O(\log n)$
 - C) $O(1)$
 - D) $O(n^2)$
-

Q.27 [2 Marks] [MODERATE] [COA — Addressing Modes]

Normal form where all non-prime attributes depend on key:

- A) 1NF
 - B) 2NF
 - C) 3NF
 - D) BCNF
-

Q.28 [2 Marks] [MODERATE] [Digital Logic — Minimization]

In deadlock prevention, hold-and-wait is prevented by:

- A) Preemption

B) Resource ordering

C) Mutual exclusion

D) Allocation at once

Q.29 [2 Marks] [MODERATE] [Engineering Mathematics — Numerical Methods]

Class A IP range:

- A) 0-127
 - B) 128-191
 - C) 192-223
 - D) 224-239
-

Q.30 [2 Marks] [MODERATE] [Programming — Recursion]

A regular grammar generates:

- A) Type 0
 - B) Type 1
 - C) Type 2
 - D) Type 3
-

Q.31 [2 Marks] [MODERATE] [Algorithms — Greedy]

Cache hit ratio if access time = 10ns, hit time = 1ns, miss penalty = 100ns:

- A) 0.1
 - B) 0.5
 - C) 0.9
 - D) 0.99
-

Q.32 [2 Marks] [MODERATE] [Data Structures — Heaps]

Which is not a page replacement algorithm?

- A) FIFO
 - B) LRU
 - C) Optimal
 - D) FCFS
-

Q.33 [2 Marks] [HARD] [DBMS — Indexing]

Which data structure implements FIFO?

- A) Stack
 - B) Queue
 - C) Tree
 - D) Graph
-

Q.34 [2 Marks] [HARD] [OS — File Systems]

Time complexity of binary search?

- A) $O(n)$
 - B) $O(\log n)$
 - C) $O(1)$
 - D) $O(n^2)$
-

Q.35 [2 Marks] [HARD] [CN — Routing]

Normal form where all non-prime attributes depend on key:

- A) 1NF

B) 2NF

C) 3NF

D) BCNF

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Section B — Multiple Select Questions (MSQ)

Q.36 [2 Marks] [HARD] [TOC — Pushdown Automata]

Which are NP-Complete problems?

- A) Hamiltonian Cycle
 - B) Sorting
 - C) Clique
 - D) Binary Search
 - E) Boolean SAT
-

Q.37 [2 Marks] [HARD] [COA — Instruction Formats]

Which are stable sorting algorithms?

- A) Bubble
 - B) Insertion
 - C) Merge
 - D) Quick
 - E) Selection
-

Q.38 [2 Marks] [HARD] [Digital Logic — Combinational Circuits]

Which are valid lossless join decompositions?

- A) BCNF
 - B) 3NF
 - C) 2NF
 - D) 1NF
 - E) All always lossless
-

Q.39 [2 Marks] [HARD] [Engineering Mathematics — Numerical Methods]

TCP provides:

- A) Connection-oriented
 - B) Reliable
 - C) Unreliable
 - D) Flow control
 - E) Error recovery
-

Q.40 [2 Marks] [HARD] [Programming — Recursion]

Which are NP-Complete problems?

- A) Hamiltonian Cycle
 - B) Sorting
 - C) Clique
 - D) Binary Search
 - E) Boolean SAT
-

Section C — Numerical Answer Type (NAT)**Q.51 [2 Marks] [HARD] [Algorithms — Greedy]**

Array A[1..100] traversed once. Time complexity?

Answer: _____

Q.52 [2 Marks] [HARD] [Data Structures — Linked Lists]

BST with 7 nodes at height 2 (root=height 0). Min nodes?

Answer: _____

Q.53 [2 Marks] [HARD] [DBMS — Transactions]

GATE CSE 2023: Semaphore with S=3, 5 processes need 1 each. How many wait?

Answer: _____

Q.54 [2 Marks] [HARD] [OS — File Systems]

IPv4 header length = 20 bytes without options. In 32-bit words?

Answer: _____

Q.55 [2 Marks] [HARD] [CN — Network Security]

Quick sort: partition on 10 elements. Pivot comparisons ~?

Answer: _____

Q.56 [2 Marks] [HARD] [TOC — Regular Languages]

Array A[1..100] traversed once. Time complexity?

Answer: _____

Q.57 [2 Marks] [HARD] [COA — Cache Memory]

BST with 7 nodes at height 2 (root=height 0). Min nodes?

Answer: _____

Q.58 [2 Marks] [HARD] [Digital Logic — Minimization]

GATE CSE 2023: Semaphore with S=3, 5 processes need 1 each. How many wait?

Answer: _____

Q.59 [2 Marks] [HARD] [Engineering Mathematics — Linear Algebra]

IPv4 header length = 20 bytes without options. In 32-bit words?

Answer: _____

Q.60 [2 Marks] [HARD] [Programming — Data Structures]

Quick sort: partition on 10 elements. Pivot comparisons ~?

Answer: _____

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Answer Key

Verify your answers after completing the full paper.

Q.1:	B	Q.2:	B	Q.3:	B
Q.4:	D	Q.5:	A	Q.6:	D
Q.7:	C	Q.8:	D	Q.9:	B
Q.10:	B	Q.11:	B	Q.12:	D
Q.13:	A	Q.14:	D	Q.15:	C
Q.16:	D	Q.17:	B	Q.18:	B
Q.19:	B	Q.20:	D	Q.21:	A
Q.22:	D	Q.23:	C	Q.24:	D
Q.25:	B	Q.26:	B	Q.27:	B
Q.28:	D	Q.29:	A	Q.30:	D
Q.31:	C	Q.32:	D	Q.33:	B
Q.34:	B	Q.35:	B	Q.36:	ACE
Q.37:	ABC	Q.38:	ABC	Q.39:	ABDE
Q.40:	ACE	Q.41:	ABC	Q.42:	ABC
Q.43:	ABDE	Q.44:	ACE	Q.45:	ABC
Q.46:	ABC	Q.47:	ABDE	Q.48:	ACE
Q.49:	ABC	Q.50:	ABC	Q.51:	100
Q.52:	7	Q.53:	2	Q.54:	5
Q.55:	9	Q.56:	100	Q.57:	
Q.58:	2	Q.59:	5	Q.60:	9

Detailed Solutions

Question 1 [Algorithms — Graph Algorithms]

Which data structure implements FIFO?

Answer: (B)

Queue follows First-In-First-Out (FIFO) principle.

Question 2 [Data Structures — Hashing]

Time complexity of binary search?

Answer: (B)

Binary search halves search space each step !' $O(\log n)$.

Question 3 [DBMS — Indexing]

Normal form where all non-prime attributes depend on key:

Answer: (B)

2NF: no partial dependency of non-prime attributes on key.

Question 4 [OS — File Systems]

In deadlock prevention, hold-and-wait is prevented by:

Answer: (D)

Allocation at once (request all resources at beginning) prevents hold-and-wait.

Question 5 [CN — TCP/IP]

Class A IP range:

Answer: (A)

Class A: 0.0.0.0 - 127.255.255.255. First octet 0-127.

Question 6 [TOC — Turing Machines]

A regular grammar generates:

Answer: (D)

Regular grammars (Type 3) generate regular languages.

Question 7 [COA — Addressing Modes]

Cache hit ratio if access time = 10ns, hit time = 1ns, miss penalty = 100ns:

Answer: (C)

AMAT = Hit time + Miss rate $\times$ Miss penalty = 1 + (1-h) $\times$ 100 = 10 !' h = 0.9.

Question 8 [Digital Logic — Combinational Circuits]

Which is not a page replacement algorithm?

Answer: (D)

FCFS is CPU scheduling. FIFO, LRU, Optimal are page replacement.

Question 9 [Engineering Mathematics — Linear Algebra]

Which data structure implements FIFO?

Answer:

(B)

Queue follows First-In-First-Out (FIFO) principle.

Question 10 [Programming — C Programming]

Time complexity of binary search?

Answer: (B)

Binary search halves search space each step !' $O(\log n)$.

Question 11 [Algorithms — Time Complexity]

Normal form where all non-prime attributes depend on key:

Answer: (B)

2NF: no partial dependency of non-prime attributes on key.

Question 12 [Data Structures — Hashing]

In deadlock prevention, hold-and-wait is prevented by:

Answer: (D)

Allocation at once (request all resources at beginning) prevents hold-and-wait.

Question 13 [DBMS — Transactions]

Class A IP range:

Answer: (A)

Class A: 0.0.0.0 - 127.255.255.255. First octet 0-127.

Question 14 [OS — CPU Scheduling]

A regular grammar generates:

Answer: (D)

Regular grammars (Type 3) generate regular languages.

Question 15 [CN — Routing]

Cache hit ratio if access time = 10ns, hit time = 1ns, miss penalty = 100ns:

Answer: (C)

$AMAT = \text{Hit time} + \text{Miss rate} \times \text{Miss penalty} = 1 + (1-h) \times 100 = 10$!' $h = 0.9$.

Question 16 [TOC — Grammars]

Which is not a page replacement algorithm?

Answer: (D)

FCFS is CPU scheduling. FIFO, LRU, Optimal are page replacement.

Question 17 [COA — Addressing Modes]

Which data structure implements FIFO?

Answer: (B)

Queue follows First-In-First-Out (FIFO) principle.

Question 18 [Digital Logic — Minimization]

Time complexity of binary search?

Answer:

(B)

Binary search halves search space each step !' $O(\log n)$.

Question 19 [Engineering Mathematics — Calculus]

Normal form where all non-prime attributes depend on key:

Answer: (B)

2NF: no partial dependency of non-prime attributes on key.

Question 20 [Programming — Data Structures]

In deadlock prevention, hold-and-wait is prevented by:

Answer: (D)

Allocation at once (request all resources at beginning) prevents hold-and-wait.

Question 21 [Algorithms — Greedy]

Class A IP range:

Answer: (A)

Class A: 0.0.0.0 - 127.255.255.255. First octet 0-127.

Question 22 [Data Structures — Hashing]

A regular grammar generates:

Answer: (D)

Regular grammars (Type 3) generate regular languages.

Question 23 [DBMS — Transactions]

Cache hit ratio if access time = 10ns, hit time = 1ns, miss penalty = 100ns:

Answer: (C)

AMAT = Hit time + Miss rate $\times$ Miss penalty = 1 + (1-h) $\times$ 100 = 10 !' h = 0.9.

Question 24 [OS — File Systems]

Which is not a page replacement algorithm?

Answer: (D)

FCFS is CPU scheduling. FIFO, LRU, Optimal are page replacement.

Question 25 [CN — Network Security]

Which data structure implements FIFO?

Answer: (B)

Queue follows First-In-First-Out (FIFO) principle.

Question 26 [TOC — Decidability]

Time complexity of binary search?

Answer: (B)

Binary search halves search space each step !' $O(\log n)$.

Question 27 [COA — Addressing Modes]

Normal form where all non-prime attributes depend on key:

Answer:

(B)

2NF: no partial dependency of non-prime attributes on key.

Question 28 [Digital Logic — Minimization]

In deadlock prevention, hold-and-wait is prevented by:

Answer: (D)

Allocation at once (request all resources at beginning) prevents hold-and-wait.

Question 29 [Engineering Mathematics — Numerical Methods]

Class A IP range:

Answer: (A)

Class A: 0.0.0.0 - 127.255.255.255. First octet 0-127.

Question 30 [Programming — Recursion]

A regular grammar generates:

Answer: (D)

Regular grammars (Type 3) generate regular languages.

Question 31 [Algorithms — Greedy]

Cache hit ratio if access time = 10ns, hit time = 1ns, miss penalty = 100ns:

Answer: (C)

AMAT = Hit time + Miss rate $\times$ Miss penalty = $1 + (1-h) \times 100 = 10$!' $h = 0.9$.

Question 32 [Data Structures — Heaps]

Which is not a page replacement algorithm?

Answer: (D)

FCFS is CPU scheduling. FIFO, LRU, Optimal are page replacement.

Question 33 [DBMS — Indexing]

Which data structure implements FIFO?

Answer: (B)

Queue follows First-In-First-Out (FIFO) principle.

Question 34 [OS — File Systems]

Time complexity of binary search?

Answer: (B)

Binary search halves search space each step !' $O(\log n)$.

Question 35 [CN — Routing]

Normal form where all non-prime attributes depend on key:

Answer: (B)

2NF: no partial dependency of non-prime attributes on key.

Question 36 [TOC — Pushdown Automata]

Which are NP-Complete problems?

Answer:

(A, C, E)

Hamiltonian Cycle, Clique, SAT are NP-Complete. Sorting, Binary Search are P.

Question 37 [COA — Instruction Formats]

Which are stable sorting algorithms?

Answer: (A, B, C)

Bubble, Insertion, Merge are stable. Quick, Selection are not.

Question 38 [Digital Logic — Combinational Circuits]

Which are valid lossless join decompositions?

Answer: (A, B, C)

Decomposition to BCNF may not preserve dependencies. 3NF is always lossless.

Question 39 [Engineering Mathematics — Numerical Methods]

TCP provides:

Answer: (A, B, D, E)

TCP is connection-oriented, reliable, has flow control, error recovery.

Question 40 [Programming — Recursion]

Which are NP-Complete problems?

Answer: (A, C, E)

Hamiltonian Cycle, Clique, SAT are NP-Complete. Sorting, Binary Search are P.

Question 41 [Algorithms — Dynamic Programming]

Which are stable sorting algorithms?

Answer: (A, B, C)

Bubble, Insertion, Merge are stable. Quick, Selection are not.

Question 42 [Data Structures — Hashing]

Which are valid lossless join decompositions?

Answer: (A, B, C)

Decomposition to BCNF may not preserve dependencies. 3NF is always lossless.

Question 43 [DBMS — Indexing]

TCP provides:

Answer: (A, B, D, E)

TCP is connection-oriented, reliable, has flow control, error recovery.

Question 44 [OS — CPU Scheduling]

Which are NP-Complete problems?

Answer: (A, C, E)

Hamiltonian Cycle, Clique, SAT are NP-Complete. Sorting, Binary Search are P.

Question 45 [CN — Network Security]

Which are stable sorting algorithms?

Answer:

(A, B, C)

Bubble, Insertion, Merge are stable. Quick, Selection are not.

Question 46 [TOC — Turing Machines]

Which are valid lossless join decompositions?

Answer: (A, B, C)

Decomposition to BCNF may not preserve dependencies. 3NF is always lossless.

Question 47 [COA — I/O]

TCP provides:

Answer: (A, B, D, E)

TCP is connection-oriented, reliable, has flow control, error recovery.

Question 48 [Digital Logic — ADC/DAC]

Which are NP-Complete problems?

Answer: (A, C, E)

Hamiltonian Cycle, Clique, SAT are NP-Complete. Sorting, Binary Search are P.

Question 49 [Engineering Mathematics — Numerical Methods]

Which are stable sorting algorithms?

Answer: (A, B, C)

Bubble, Insertion, Merge are stable. Quick, Selection are not.

Question 50 [Programming — OOP]

Which are valid lossless join decompositions?

Answer: (A, B, C)

Decomposition to BCNF may not preserve dependencies. 3NF is always lossless.

Question 51 [Algorithms — Greedy]

Array $A[1..100]$ traversed once. Time complexity?

Answer: (100)

Traversing n elements !' $O(n) = 100$ operations.

Question 52 [Data Structures — Linked Lists]

BST with 7 nodes at height 2 (root=height 0). Min nodes?

Answer: (7)

Full binary tree at height 2 has 7 nodes (perfect binary tree).

Question 53 [DBMS — Transactions]

GATE CSE 2023: Semaphore with $S=3$, 5 processes need 1 each. How many wait?

Answer: (2)

3 proceed immediately. 2 wait. Total blocked = 2.

Question 54 [OS — File Systems]

IPv4 header length = 20 bytes without options. In 32-bit words?

Answer:

20 bytes / 4 bytes per word = 5 words.

Question 55 [CN — Network Security]

Quick sort: partition on 10 elements. Pivot comparisons ~?

Answer: (9)

Partitioning n elements requires $n-1$ comparisons. For 10: 9.

Question 56 [TOC — Regular Languages]

Array $A[1..100]$ traversed once. Time complexity?

Answer: (100)

Traversing n elements !' $O(n) = 100$ operations.

Question 57 [COA — Cache Memory]

BST with 7 nodes at height 2 (root=height 0). Min nodes?

Answer: (7)

Full binary tree at height 2 has 7 nodes (perfect binary tree).

Question 58 [Digital Logic — Minimization]

GATE CSE 2023: Semaphore with $S=3$, 5 processes need 1 each. How many wait?

Answer: (2)

3 proceed immediately. 2 wait. Total blocked = 2.

Question 59 [Engineering Mathematics — Linear Algebra]

IPv4 header length = 20 bytes without options. In 32-bit words?

Answer: (5)

20 bytes / 4 bytes per word = 5 words.

Question 60 [Programming — Data Structures]

Quick sort: partition on 10 elements. Pivot comparisons ~?

Answer: (9)

Partitioning n elements requires $n-1$ comparisons. For 10: 9.
